

### Interpersonal Questions 10 February

- Tell me about yourself.
- What are your greatest strengths?
- What are your greatest weaknesses?
- Why should I hire you?
- Why do you want to work at our company?
- Where do you see yourself five years from now?
- What is your salary expectation?
- Do you have any questions for us?

### 11 February

- What is the difference between Block-level and Inline elements in HTML?
- What are Semantic HTML elements, and why are they important?
- What are HTML Entities, and when would you use them?
- What is the purpose of the <meta> tag in the HTML <head> section, and what are some important attributes it can have?
- What are Void/Empty elements in HTML? Provide an example.
- Explain the CSS Box Model. What are its components?
- What are the three ways to include CSS in an HTML document, and which one is generally preferred?
- Explain the difference between id and class selectors in CSS.
- What is CSS Specificity, and what is the general order of precedence when styles conflict?
- What is the difference between display: none; and visibility: hidden; in CSS?

### 12 February

- What is the difference between var, let, and const in JavaScript?
- How does the JavaScript event loop work?
- Explain closures in JavaScript with an example.
- What are promises, and how do they differ from async/await?
- What is the virtual DOM, and how does React use it?

- Explain the difference between Flexbox and CSS Grid. When would you choose one over the other?
- What is the purpose of CSS Media Queries, and how do you use them to create a responsive design?
- Explain the CSS `position` property and differentiate between the values `relative`, `absolute`, `fixed`, and `sticky`.
- How does the `this` keyword work in JavaScript, and how do arrow functions change its behavior?
- What are the key differences between a React Class Component and a Functional Component with Hooks? Which one is generally preferred today and why?

### 13 February

- What is the purpose of the `z-index` property, and what conditions must be met for it to take effect?
- What are CSS Pseudo-classes and Pseudo-elements? Provide an example of a common use case for each.
- How do you horizontally and vertically center an element using Flexbox or CSS Grid? Provide a code example for one method.
- What is hoisting in JavaScript? Explain how `var`, `let`, and `const` variables are affected by it.
- Differentiate between Event Bubbling and Event Capturing. Which one does modern JavaScript event handlers use by default?
- Explain the difference between `==` and `===` in JavaScript. When does type coercion occur?
- Explain the purpose of the `useEffect` Hook in React. What is the role of the dependency array, and what happens if you omit it?
- What is "Prop Drilling," and what are the two most common ways to avoid it in React?
- Explain the difference between `useState` and `useReducer`. When should you choose `useReducer` over `useState`?
- What is a Higher-Order Component (HOC), and how does its use compare to Custom Hooks?
- How can you optimize a React Functional Component to prevent unnecessary re-renders? Discuss `React.memo` and the `useCallback` Hook.

#### 14 February

- What is the difference between Local Storage, Session Storage, and Cookies? When should you use each one?
- What is the difference between `useMemo` and `useCallback`? Provide a scenario where each hook is necessary for optimization.
- Explain the Context API in React.
- Describe the purpose of Custom Hooks in React. How do you create one, and what rule must they always follow?
- Explain the concept of JavaScript Closures. How do they affect variable scope and provide a practical use case, such as creating private methods?
- What is Express.js, and how does it work?
- What are middlewares in Express.js?
- How do you handle errors in Express.js applications?
- What is Node.js, and why is it used in the MERN Stack?
- What is npm, and why is it important in MERN Stack development?

#### 15 February

- What in React are Higher-Order Components (HOC)?
- What is meant by "Callback" in Node.js?
- How does Node.js handle Child Threads?
- What are some features of MongoDB?
- What is prop drilling?
- Why use Express.js over Node.js?
- What is a RESTful API?
- Explain the event loop in Node.js?
- What is middleware in Node.js, and how is it used?
- Explain the lifecycle methods of components.

#### 16 February

- What is ReactJS? What are the advantages and disadvantages of ReactJS?
- Explain the building blocks of React?
- What distinguishes a class component from a functional component?
- What is JSX in ReactJS?
- What is a React Router? Why do we need a React Router?
- What are API Routes?
- How Next.js Works?

- What do you mean by SSR?
- How does Next.js handle client-side navigation?
- Explain the concept of dynamic routing in [Next.js](#):

17 February

- What is meant by Styled JSX in Next.js?
- Difference between the pre-rendering types available in Next.js.
- What is client-side rendering, and how does it differ from server-side rendering?
- What is the purpose of the public folder in Next.js? How do you serve static files like images, fonts, and icons?
- What is Hot Module Replacement (HMR) in Next.js, and how does it improve the development experience?
- What is the difference between layouts and templates in Next.js App Router?
- What is the difference between `getServerSideProps` & `getStaticProps` functions in Next.js?
- What is the purpose of the `getStaticPaths` function in Next.js?
- What is the purpose of the `useEffect` hook in React, and how does it relate to Next.js?
- How do you handle data fetching in Next.js?

18 February

- What are the different options for styling Next.js apps?
- How do you work with custom server middleware in Next.js?
- Explain the purpose of the `_app.js` file in Next JS.
- Explain the concept of "Serverless" deployment in the context of Next JS. How does it work, and what are the advantages?
- Why use Create Next App?
- What are Environment Variables in Next.js?

- Explain the concept of "prefetching" in Next.js and how it impacts performance.
- How can you handle cross-origin requests (CORS) in a Next.js application when making API requests to a different domain?

19 February

- What is serverless architecture, and how does it relate to Next.js?
- How do you optimize the performance of a Next.js application?
- How would you optimize the performance of a large-scale Next.js application?
- How can you implement authentication in Next.js? Compare JWT, NextAuth, and Firebase approaches.
- Explain the MVC architecture?
- What are some features of MongoDB?
- Why use Express.js over Node.js?